Language-Guided Time-Series Event Detection

with Zero-Shot Cross-Modal Grounding

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FINDINGS OF EMNLP

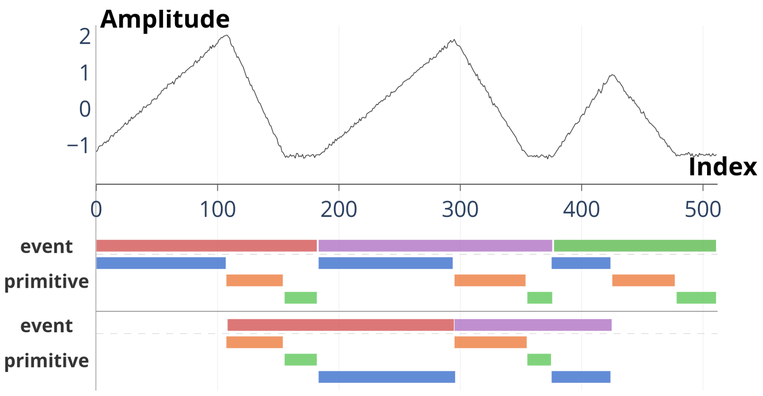
Describe an event in natural language. GroundingMOMENT returns every occurrence and its exact temporal structure—without domain-specific fine-tuning.

Why language-guided detection?

Conventional classifiers label a fixed window; they do not tell us **where an event starts and ends**. Domain-specific detectors also need scarce expert annotations.

Domain expertise already exists in manuals, textbooks, and natural-language descriptions.

Task: given a time series **x** and description Q, predict event spans (s, e) and the ordered primitive spans that compose each event.



Query order changes the detections. The same signal is grounded differently for the canonical and permuted descriptions, showing that the model captures sequential structure rather than only shape.

Three contributions

1. **GroundingMOMENT**: a fast, hierarchical model for joint event and primitive span detection.
2. **GroundTSSyn**: a scalable pipeline for synthetic, perfectly localized time-series grounding data.
3. **GroundTSBench**: the first open benchmark for language-guided time-series grounding.

GroundTSBench at a glance

16,410 time-series samples

8.8 descriptions / sample

4.4 events / sample

4.7 primitives / event

Six structural primitives:

increase	decrease	plateau
spike	dip	oscillation

Perfect boundaries: spans are derived programmatically. Language models vary only the surface wording; a critic filters descriptions.

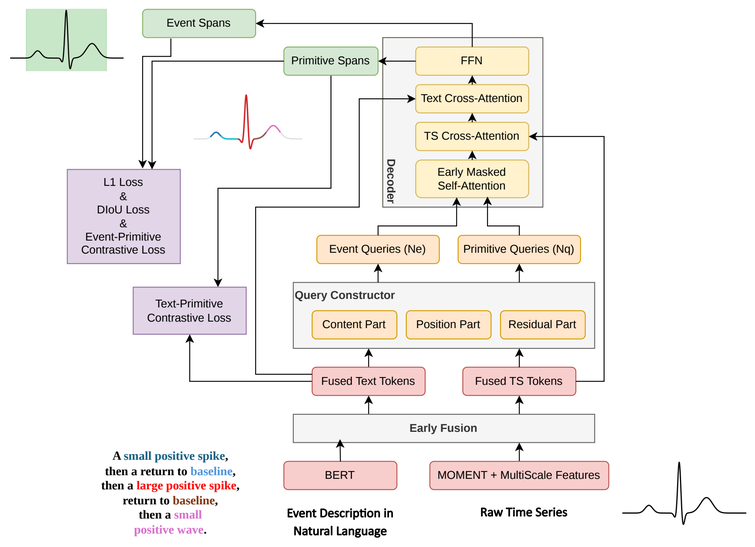
Synthetic data, real diversity

PIPELINE

Layout planning → primitive synthesis → normalization & noise → bounded occurrence jitter → descriptions & hard negatives → critic review.

Periodic, baseline-bounded, and free-form signals expose the model to varied morphologies while preserving exact ground truth.

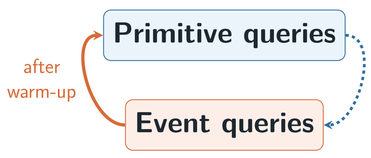
GroundingMOMENT



A DETR-style temporal grounding architecture adapted from open-vocabulary visual grounding to 1D time series.

1. **Encode at multiple scales.** MOMENT turns 512 samples into a hierarchy of temporal tokens; BERT encodes the event description.
2. **Fuse early.** Bidirectional cross-attention enriches time-series tokens with linguistic intent and text tokens with signal context.
3. **Initialize with language.** Primitive queries select phrase-relevant tokens; event queries use a pooled whole-description match.
4. **Decode jointly.** Event and primitive queries predict (start, length) spans in one hierarchical decoder.

Compositional attention is the key



Warm-up: isolate event queries while primitive detectors stabilize.

Thereafter: events may aggregate primitives, but primitives never shortcut through event context.

This directional constraint makes the model **compose events from domain-agnostic structure** instead of memorizing event waveforms.

What matters? Ablation evidence

Full model	79.2
No query-class residual	71.1
No warm-up mask	73.2
No permanent mask	71.0
Event-only decoder	67.4

Event mAP on GroundTSBench. Removing primitive supervision causes the largest drop; the hierarchy is not merely an auxiliary objective.

Generalization test

Lexical holdout: train without the canonical words *spike, increase, decrease, plateau, oscillation, dip*; test only on descriptions containing them.

76.1 event mAP

87.9 primitive mAP

Close to the full model (79.2 / 88.7), indicating semantic grounding beyond memorized primitive names.

Accurate grounding at practical latency

EVENT MAP ON GROUNDTSBENCH

SELA + GPT-5.4	83.6
GroundingMOMENT	79.2
GPT-5.4 (text)	23.8
ChatTS	2.3

40× faster

1.2 s/sample vs. 49.0 s for the best agentic system

88.7 primitive mAP

77.5 event mIoU@0.7

Zero-shot transfer to ECG

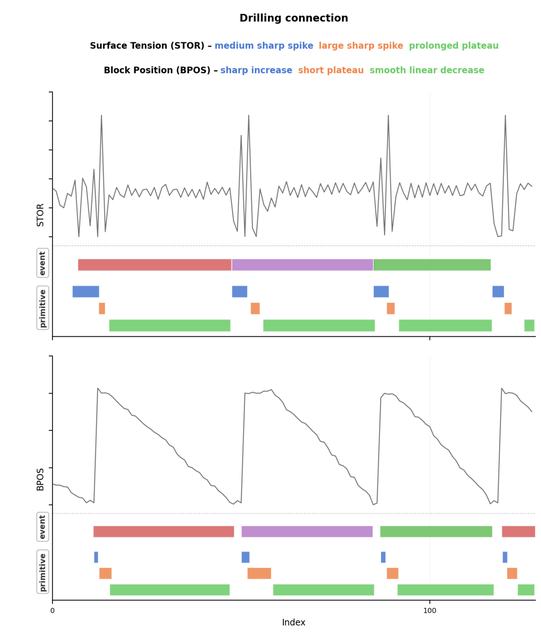


The **same AFIB segment** is queried twice. A normal-sinus description yields no events (top); replacing the missing P wave with a plateau localizes all four AFIB beats (bottom).

63.0 ECG mAP

1.3 s latency

Zero-shot transfer to drilling



Independent text queries ground connection phases in surface torque and block position. Primitive spans expose phase durations, not only event counts.

67.2 drilling mAP

2.5 s latency

Operational insight: reattachment has the highest duration variability (CV = 85.0%), making it the primary improvement target.

Flexible interface

New event descriptions can be queried without retraining a domain-specific detector.

Precise & auditable

Joint event and primitive spans expose the structural evidence behind each detection.

Deployment-ready speed

Near-agentic accuracy without iterative tool calls or per-query token overhead.

Take-away: synthetic primitives + natural language unlock fast, auditable, zero-shot event localization across domains.

Code, benchmark, and paper: github.com/houtj/GroundingMOMENT

Scan for code

GroundTSBench is linked from the repository.